# Onatola Timilehin Faruq

**Software Engineer | Blockchain Wallet Infrastructure | Backend Systems**
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## Summary

Backend-focused blockchain engineer with production experience building wallet infrastructure, smart contract systems, and scalable server-side applications. Built multi-chain HD wallets (BIP-39/32/44), privacy-preserving payment protocols, and NFT marketplaces handling 1K+ concurrent users. Deep understanding of digital asset security: key derivation, stealth addresses, access control patterns, and automated asset management, directly relevant to Cache Wallet's mission of safeguarding and transferring digital assets.

## Technical Skills

**Languages:** TypeScript, Solidity, Rust, Python, Noir, Clarity

**Backend:** Node.js, Express, PostgreSQL, Redis, WebSocket, GraphQL, REST APIs

**Blockchain:** Ethers.js, Viem, Wagmi, Foundry, Hardhat, Slither, The Graph

**Frontend:** React, Next.js, Tailwind CSS

**Infrastructure:** Docker, GitHub Actions, IPFS, Arweave, Tenderly

**Security:** Smart contract auditing, fuzzing, formal verification, access control, key management

**Chains:** Ethereum, Base, Arbitrum, Solana, Hedera, Stacks, BNB Chain

## Professional Experience

### Blockchain Engineer | DeFiConnectCredit | Remote | Jan 2025 - Present

- Architected modular smart contract systems integrating Aave v3, Uniswap v3, GMX, and Curve with 99.9% uptime
- Built reusable contract libraries reducing partner integration time by 60%
- Established Foundry testing framework with 95%+ coverage, catching 3 critical vulnerabilities pre-deployment
- Reduced transaction costs by 35% through assembly optimization and storage pattern improvements
- Led API/SDK development for external partner integrations with comprehensive documentation

### Blockchain Development Engineer | DeFiConnectCredit | Remote | Jun 2024 - Dec 2024

- Implemented core lending protocol features: collateral management, liquidation mechanics, interest rate models
- Conducted security reviews on 15+ contracts, identifying reentrancy, front-running, and oracle manipulation vulnerabilities
- Deployed ERC721 incentive system with Merkle tree proofs serving 500+ users
- Achieved 90%+ test coverage including edge case and stress testing

## Relevant Projects

### OmniShield Wallet: Multi-Chain HD Wallet

*TypeScript, React, React Native | github.com/Leihyn/omnishield-wallet*

Multi-chain wallet supporting 6 blockchains with privacy-first design. Directly relevant to Cache Wallet's non-custodial wallet architecture.

- Implemented BIP-39/32/44 key derivation across Zcash, Miden, Aztec, Mina, Fhenix, Osmosis
- Built chain-specific adapters with unified interface for balance queries, transactions, and signing
- Created privacy scoring algorithm with auto-shield recommendations
- Developed both React web and React Native mobile interfaces
- Integrated with OmniSwap SDK for cross-chain swap functionality

### Privacy Chronicles: Glass Wallet (Winner, Solana Privacy Hackathon 2026)

*JavaScript, Next.js, React Three Fiber, Solana | github.com/Leihyn/glass-wallet-comic | Live Demo*

Won Solana Privacy Hackathon building an interactive comic series that teaches blockchain wallet privacy (MEV protection, surveillance resistance, stealth addresses) through narrative. Directly maps to Cache Wallet's mission.

- 12-page interactive comic on selective wallet privacy (MEV, surveillance, stealth addresses) for Solana
- 5-episode series (70+ pages) covering zero-knowledge proofs with 3D solar system visualization in React Three Fiber
- Also won Content Creation Track at Zypherpunk Hackathon (Mina Foundation)

### Nocturne: Private Payments on Solana

*Rust, Anchor, Noir, Zero-Knowledge | github.com/Leihyn/nocturne*

Privacy protocol with secure asset transfer mechanics. Relevant to Cache Wallet's decentralized asset transfer/will functionality.

- Implemented stealth address system (DKSAP) for recipient privacy in asset transfers
- Built fixed-denomination privacy pools with Merkle commitments achieving 97% unlinkability
- Created ZK proof circuits (Noir/UltraHonk) for withdrawal verification
- Developed TEE relay system for metadata protection

### Comic Pad: Production NFT Marketplace

*Hedera, Next.js, Node.js, PostgreSQL, Redis | github.com/Leihyn/comicPad*

Full-stack platform demonstrating scalable backend architecture and database management.

- Solo-built complete platform: smart contracts, REST API, WebSocket server, frontend

- Engineered backend handling 10K+ daily requests with <100ms response time
- Built distributed job queue (Bull/Redis) processing 500+ operations/hour
- Created WebSocket server delivering real-time updates to 1K+ concurrent users (<50ms latency)
- Implemented hybrid IPFS/Arweave storage with automatic failover and 99.9% availability

**OmniSwap SDK: Cross-Chain Swap Middleware**

*TypeScript | github.com/Leihyn/omniswap-sdk*

Production SDK demonstrating system design, OOP patterns, and robust error handling.

- Architected chain-agnostic adapter pattern (OOP) supporting 6 heterogeneous blockchains
- Implemented HTLC-based atomic swap protocol with automated timeout and refund management
- Built typed error hierarchy, exponential backoff retry logic, and circuit breakers
- Created 70KB+ documentation with API reference, tutorials, and working examples

**Sentinel: AI-Powered Smart Contract Auditor**

*Python, Slither, Foundry | github.com/Leihyn/sentinel*

Multi-agent security framework relevant to Cache Wallet's need for secure, reliable systems.

- Built automated auditing pipeline: static analysis, vulnerability hunting, PoC generation
- Detects oracle manipulation, flash loan attacks, reentrancy, and access control issues
- Generates Foundry-based proof-of-concept exploits with mainnet fork validation

## Achievements

- **Winner:** Solana Privacy Hackathon 2026 (Privacy Chronicles: Glass Wallet)
- **Content Track Winner:** Zypherpunk Hackathon (Mina Foundation) (Privacy Chronicles)
- **2nd Place:** Seedify Prediction Market Hackathon (TruthBounty)
- **Graduate:** Uniswap Hook Incubator (UHI7), 15% acceptance rate
- **Certified:** Cyfrin Updraft (DeFi Security & Protocol Engineering)
- **Certified:** Hashgraph Developer (The Hashgraph Association)

## Education

**Bachelor of Science in Physiotherapy** | University of Ibadan | 2024

**Blockchain Engineering** (equivalent coursework): Uniswap Hook Incubator (UHI7), Cyfrin Updraft (DeFi Security), School of Solana (Ackee Blockchain), Hashgraph Developer Certification

*Portfolio: faruukku.vercel.app | GitHub: github.com/Leihyn*